Credit Default Risk Prediction

Access: https://github.com/mandypeng-brown/data1030-project-credit-risk.git

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Project Overview – Motivation

The Goal: Predict if a loan applicant will default.

Problem Type: Binary Classification (Default vs. Non-default)

Why It Matters:

- For Lenders: Reduce financial risk & losses.
- For Applicants: Increase fair access to credit.

Project Overview – Data & Challenge

The Data:

- **Source**: Kaggle's "Home Credit Default Risk" Competition.
- **Content**: Application data + Historical loan records.

application_dataset.csv
Info about loan and applicant at application time.
One row represents one loan in our data sample.
Target Variable: Target (binary).

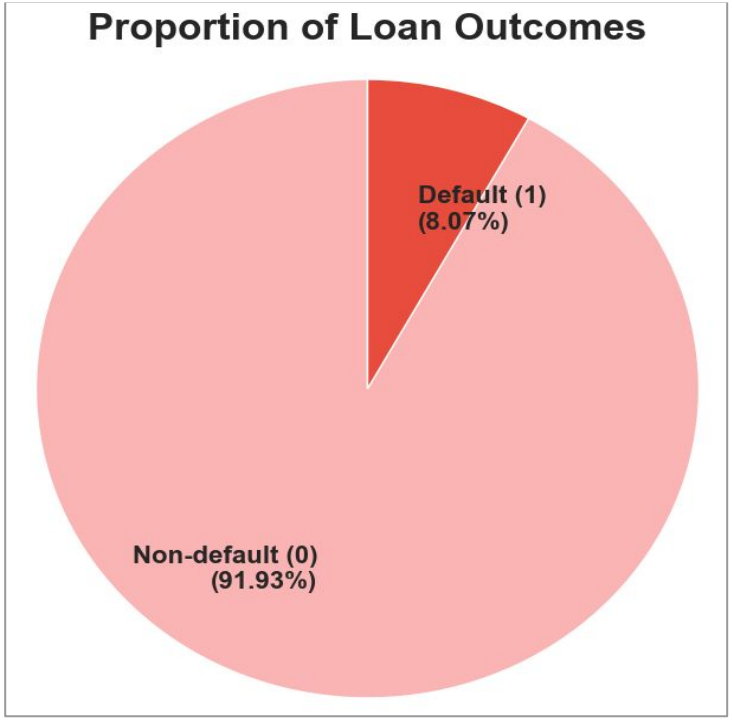
Merge and Aggregate

previous_application.csv
Info about clients' previous loans.
One row per previous application.
Foreign Key: SK_ID_CURR

Key Challenges:

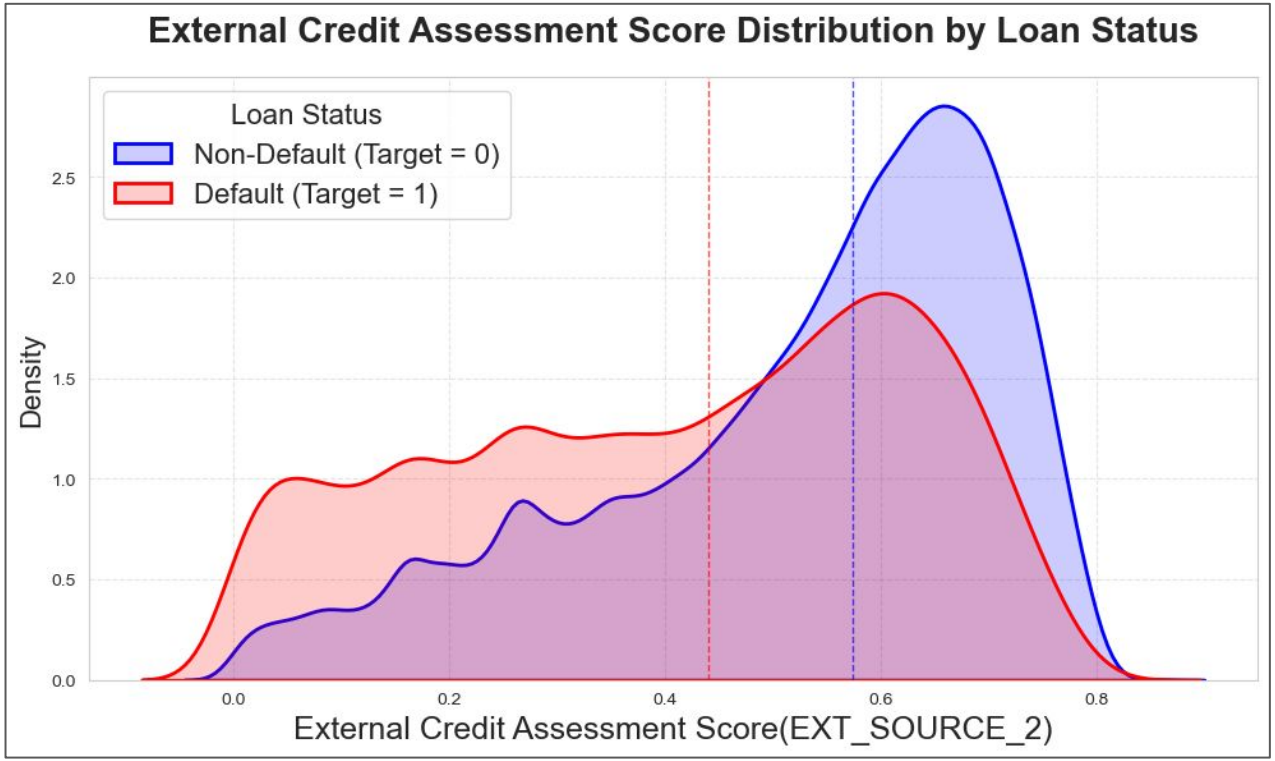
- **Large Data**: Over 300,000 applicants and 134 features.
- **Missing Data**: 50% features have missing values.

EDA Highlights – Core Challenge: Class Imbalance



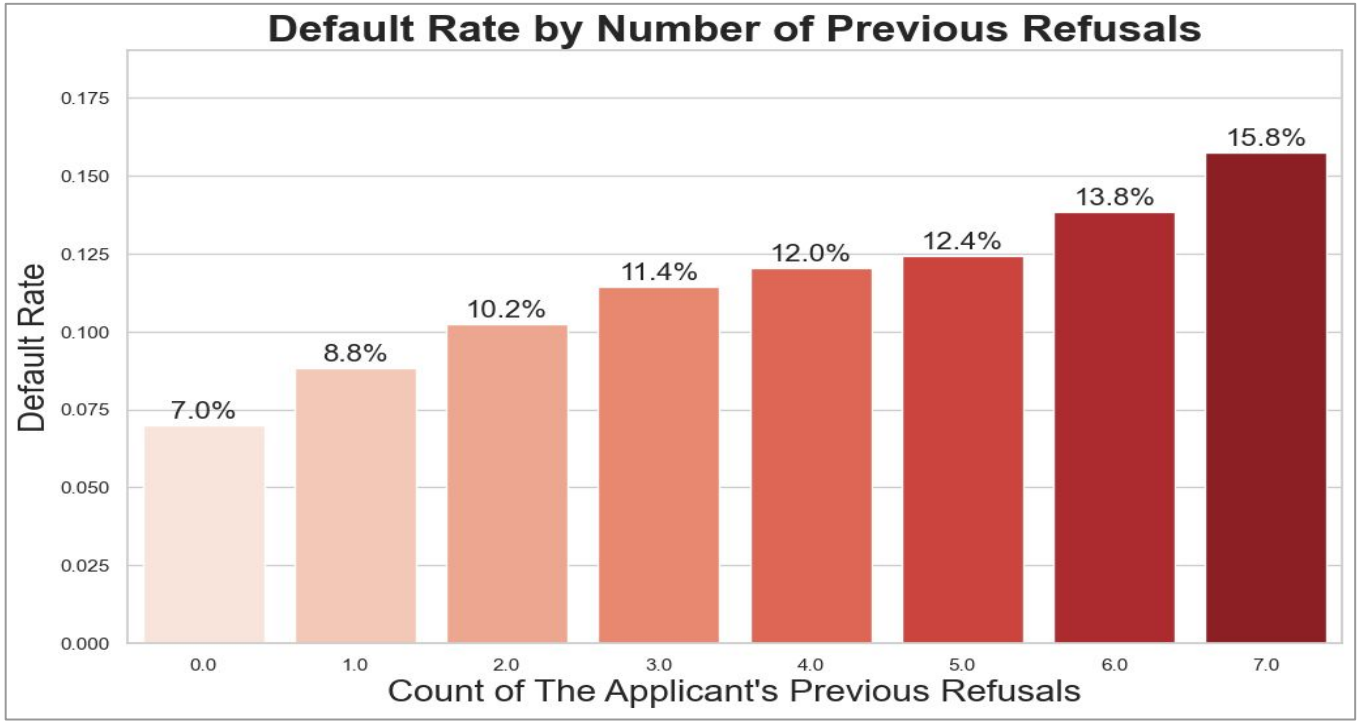
- **Finding**: Default is a rare event.
- **Implication**: This dictates our choice of metric and stratified splitting strategy.

EDA Highlights – Trustworthy External Assessment



- **Finding**: Defaulters show significantly lower scores than non-defaulters.
- **Implication**: External score is a reliable risk benchmark and strong predictor.

EDA Highlights – Power of Feature Engineering



- **Finding**: Applicants with a history of refusal are more likely to default.
- **Implication**: Merging historical data is critical for predictive power.

Data Splitting Strategy

A Two-Stage Approach for Reliable Evaluation:

- **Stage 1: Hold-out Test Set**
  - Isolate **20%** of the data for final, unbiased testing.
- **Stage 2: Cross-Validation Set**
  - All experiments are performed on the remaining **80%** using **Stratified 5-Fold Cross-Validation** for robust model evaluation.

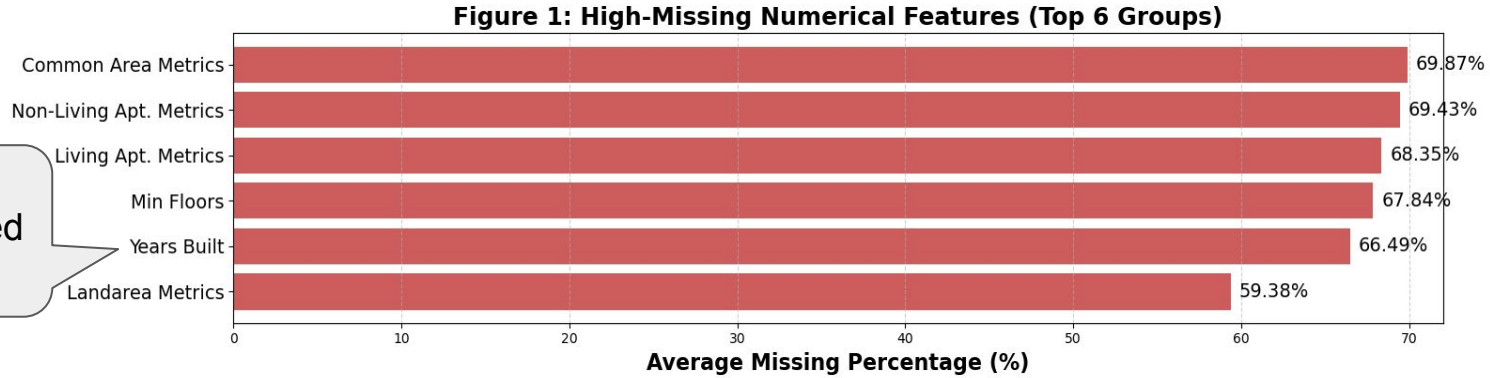
➢ **Resulting Data Partitions:**

- **Training & CV Set (80%):** 246,009 applicants | 134 features
- **Hold-out Test Set (20%):** 61,503 applicants | 134 features

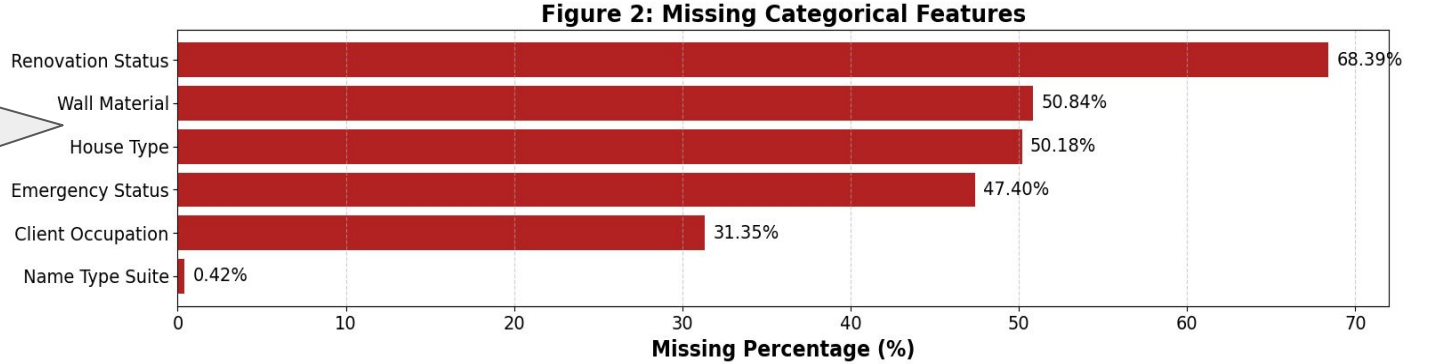
Missing Value Analysis

Summary: 67 (50%) features have missing values. Over 40 features miss >50% data.

Clustered in property-related features.



Create a new, distinct category: "Unknown".



Preprocessing Strategy

Framework: A unified scikit-learn Pipeline to prevent data leakage.

- **Transformation & Encoding:**
  - **Log Transformer**: Applied to skewed financial data (e.g., ‘Income’).
  - **Encoding**: OneHot for nominal features (e.g., ‘Occupation Type’); Ordinal for features with a hierarchy (e.g., ‘Education Level’).
  - **Standard Scaler**: Applied to numerical features after transformation.

- **The Result Dataset:** Before: 134 Features -> After: 255 Features.

Thank You for Listening!

Any Questions?